

# Sidhant Maloo

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**Junior at Huntley High School** (3.97 unweighted GPA) with a written and publicly posted machine learning paper, independent MIT OpenCourseWare study, and simulation work spanning quantum computing, black holes, and robot control. Seeking research opportunities in quantum computing, physical AI, and robotics.

## RESEARCH

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### SVR vs. Random Forest on Volatile, Noisy Stock Data *Python*

Completed 2026

- Designed and ran a comparison of Support Vector Regression and Random Forest models to see how each holds up as noise and volatility increase; wrote the full paper.
- Preprint posted on Zenodo ([zenodo.org/records/20818947](https://zenodo.org/records/20818947)); code on GitHub. Not yet peer reviewed; actively seeking expert feedback.

## COMPUTATIONAL PROJECTS

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### Interactive Simulations *six completed*

2026

- Built six simulations spanning quantum computing, black hole dynamics, robot-arm kinematics, neural networks, pathfinding, and circuit physics, implementing the underlying models directly (for example, computing qubit entanglement from the reduced density matrix).
- Code and interactive demos are public on his website and GitHub.

### Humanoid Robot ("Bob") *Python*

May 2026 – Present

- Designing an AI-controlled upper torso (two 3-DOF arms; camera, microphone, speaker) that acts on natural spoken requests.

## COURSEWORK & INDEPENDENT STUDY

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|            |                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIT OCW    | 18.01 Single Variable Calculus (completed); 18.06 Linear Algebra (in progress). Next: algorithms (6.006, 6.046J), theory of computation (6.045J), quantum physics and computation (8.04, 18.435J) |
| HHS        | AP Calculus BC and Statistics (2026–27), PLTW Digital Electronics, Honors Chemistry                                                                                                               |
| AP / Tests | Computer Science A (5), Computer Science Principles (5), Precalculus (5), Physics 1 (4); perfect PreACT 9 and 10                                                                                  |

## TECHNICAL SKILLS

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|-------------|--------------------------------------------------------------------------------------------|
| Programming | Python, Java, JavaScript, HTML/CSS, Qiskit; SVR, Random Forest, neural networks (backprop) |
| Engineering | PCB design (KiCad), Multisim, sensor systems, VEX V5, mechanical design, Raspberry Pi      |

## RELEVANT EXPERIENCE

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- VEX V5 Robotics State Team** (2024–26): programmed autonomous control from GPS, vision, and color sensors; helped earn the **Think Award** for documented engineering design and analysis.
- Captain, State Math Team**: lead practices and mentor teammates; 2× regional champion, state competitor.
- Sensei, Code Ninjas** (2026–): tutor students in Java and Unity. Taught quantum computing basics to elementary students with a Qiskit demo.
- Engineering notebook**: documents his projects in a dated log of what was tried, what broke, and what changed.

## HONORS

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UN Ambassador Award, STEM outreach (2026) • Illinois House of Representatives Honors, math (2025) • Illinois Principals Association Student of the Year (2024)